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Lab 2

Network Breach Analysis

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Table of Contents

Introduction.....	1
Findings and Methodology.....	1
1. Gaining Access.....	1
2. System Analysis.....	1
2.1 - Process Analysis Table.....	1
3. Network Analysis.....	2
3.1 - Port Analysis Table.....	3
4. Browser Artifacts and Spyware Analysis.....	4
4.1 - Cookie File Analysis Table.....	4
5. Additional Program Investigation.....	5
Incident Explanation.....	6
Future Recommendations & Conclusion.....	6
References.....	7
Appendix.....	8
1. Gaining Access.....	8
• Figure 1.1: Screenshot of the Windows XP Device Booting in Safe Mode.....	8
• Figure 1.2: Screenshot of Password Being Reset.....	8
2. System Findings.....	9
• Figure 2.1: Screenshot of Suspicious Startup Processes.....	9
• Figure 2.2: Screenshot of Windows Task Manager Process Listing.....	9
3. Network Findings.....	10
• Figure 3.1: Screenshot of all Active Connections.....	10
4. Browser Artifacts and Spyware Findings.....	11
• Figure 4.1: Screenshot of the Directory for Cookies.....	11
6. Additional Program Findings.....	11
• Figure 5.1: Screenshot of Search for Cryptcat.....	11
• Figure 5.2: Screenshot of C:\inetpub\ftproot\.....	12
• Figure 5.3: Screenshot of rundll32.exe.....	12

Introduction

This lab involves performing a forensic investigation on a compromised Windows XP system. The objective is to identify signs of unauthorized activity such as suspicious processes, files, and network connections. Using forensic investigation principles we can uncover hidden malware, spyware cookies, and remote access tools. Through further analysis, we attempt to determine how the system was compromised and assess the impact of the attack to prevent future incidents from occurring.

Findings and Methodology

1. Gaining Access

For this lab I am using the win-xp-15 (IP address: 172.16.3.225) Simspace VM. The VM was locked by malicious programs. In order to gain access, I restarted the device and booted it in safe mode. See [Figure 1.1](#) in the appendix.

Next, I navigated to the *Local Users and Groups* to reset the password of the Daniel Faraday user account. See [Figure 1.2](#) in the appendix. After resetting the password, I restarted the device and was now able to login.

2. System Analysis

Upon logging into the compromised Windows XP system, multiple suspicious processes were observed automatically running at startup. See [Figure 2.1](#) for a screenshot of the suspicious startup processes. The processes in figure 2.1 are the following:

1. *C:\inetpub\ftproot\nc.exe* – This executes Netcat, a command-line utility used to establish a remote shell.
2. *C:\WINDOWS\system32\cmd.exe* – This is an interactive command prompt session operating under the compromised user account.
3. *C:\inetpub\ftproot\VNC4\winvnc4.exe* – This is Initiating an unauthorized RealVNC server, confirming that the attacker established a full graphical remote-control backdoor.

To investigate further, I opened the Windows Task Manager. This revealed numerous suspicious processes automatically running under both system and user contexts. See [Figure 2.2](#) for a screenshot of the Windows Task Manager process listing. These processes indicate that the system has been compromised by at least one malicious actor who installed multiple remote-access tools and backdoors to maintain persistent control over the host.

2.1 - Process Analysis Table

Process	User Context	Assessment	Description
nc.exe	Daniel Faraday	Malicious	Netcat tool used to create reverse shells and open backdoors. Located in <i>C:\inetpub\ftproot\</i> , suggesting it was uploaded to the FTP server
winvnc4.exe	Daniel Faraday	Malicious	RealVNC server launched from <i>C:\inetpub\ftproot\VNC4\</i> enabling unauthorized remote desktop access.
poisonivy.exe	SYSTEM	Malicious	Known Remote Access Trojan (RAT) that provides attacker C2 access.
vjgs.exe	SYSTEM	Suspicious	Unknown binary not associated with legitimate Windows components. May act as a loader or persistence agent.
svchost.exe (multiple)	SYSTEM / NETWORK SERVICE	Potentially abused	Some svchost processes are legitimate, but the presence of several instances and possible execution from non-system directories, indicates attackers disguising malware as svchost.
msmsgs.exe	Daniel Faraday	Suspicious	Legitimate Windows Messenger executable that may be leveraged for persistence or C2 messaging.
vmtoolsd.exe, VGAAuthService.exe	SYSTEM	Benign	Legitimate VMware background services.

3. Network Analysis

The next step was to analyze the network connections. To do so, I opened the command prompt and ran the command “netstat -ano.” The command lists all active TCP/UDP connections and listening ports, along with their associated process identifiers (PIDs). See [Figure 3.1](#) for a list of all the active connections. The results provide strong evidence of malicious network activity correlating directly with the malicious processes previously identified in Task Manager.

3.1 - Port Analysis Table

Local Port and State	Protocol	PID	Associated Process	Assessment
21 - (LISTENING)	TCP	1996	inetinfo.exe	Legitimate FTP service but likely exploited as initial entry vector. Malicious files such as nc.exe and winvnc4.exe were uploaded via FTP to C:\inetpub\ftproot\.
80 - (LISTENING)	TCP	1996	inetinfo.exe	Standard HTTP port; may have been probed or exploited through IIS vulnerabilities.
135 / 139 / 445 - (LISTENING)	TCP	4	System	Common Windows RPC/SMB services, but may facilitate lateral movement or remote code execution if unpatched.
666 - (LISTENING)	TCP	188	Unknown	Highly suspicious; Port 666 is often used by IRC bots, remote shells, or backdoors. Correlates with malicious binaries found earlier.
1025 / 1029 / 1435 / 4440 - (LISTENING)	TCP	192, 1340, 1996	Possibly poisonivy.exe or winvnc4.exe	Non-standard high ports, potentially linked to Poison Ivy RAT or custom remote-access listeners.

15152 - (LISTENING)	TCP	2008	Unknown - Possibly poisonivy.exe or vjgs.exe	Another high port likely configured by malware as a command-and-control (C2) listener.
5000 / 1900 - (LISTENING)	UDP	1644	svchost.exe	Normally used by UPnP/SSDP, but could be leveraged for host discovery or persistence by malware.
5900 - (LISTENING)	UDP	1644	winvnc4.exe	Default VNC remote desktop port. This confirms that the attacker established a graphical remote control via RealVNC backdoor.
172.16.3.225 Connections - (Active)	UDP	1340/ 1644	svchost.exe, winvnc4.exe	Suggests remote communication between the victim machine and an attacker-controlled host within a private subnet.

4. Browser Artifacts and Spyware Analysis

After identifying multiple backdoor processes and unauthorized network listeners, the next step was to inspect browser and user profile data. Cookies can often reveal spyware, adware, or compromised web-tracking. To find the hidden cookies, hidden files had to be enabled through: Tools → Folder Options → View → Show hidden files and folders. After enabling hidden files, I examined the *C:\Documents and Settings\Daniel Faraday\Cookies* directory. To see the contents of the directory view [Figure 4.1](#). The directory contains multiple text files, each storing browser cookies for various web services and advertising domains.

4.1 - Cookie File Analysis Table

Cookie File	Domain/Source	Assessment
daniel faraday@ad.wsod[1].txt	ad.wsod - advertising domain	Suspicious cookie that indicates tracking or adware. Possibly injected by malicious scripts or spyware.

daniel faraday@atdmt[1].txt	atdmt.com - tracking network (Microsoft Advertising)	Potential Spyware / Ad Tracking cookie commonly exploited by early-2000s adware to harvest browsing data.
daniel faraday@exp.www.msn[1].txt	msn.com	Possibly a normal MSN cookie, but could also store session data exploitable by RATs.
daniel faraday@scorecardresearch[1].txt	scorecardresearch.com - analytics network	A tracking cookie often bundled with adware or analytics spyware.
daniel faraday@sun[1].txt	sun.com - Java or browser plugin update	Legitimate and related to Java updates.
daniel faraday@openoffice[2].txt	openoffice.org	Benign, application update server.
daniel faraday@live[1].txt, @rad.msn[2].txt, @c.msn[1].txt	MSN / Microsoft Live	Likely legitimate session cookies but still sensitive; If stolen it will leak credentials.

5. Additional Program Investigation

After finding these malicious programs, further investigation was needed. The next step was to rule out the presence of any other malicious programs like cryptcat.exe (an encrypted Netcat variant) or rundll32.exe (a Windows binary commonly abused by attackers).

Two searches were performed to identify possible cryptcat installations:

1. C:\Documents and Settings\Daniel Faraday>dir /s /b c:\cryptcat.exe
2. C:\Documents and Settings\Daniel Faraday>dir /s /b *cryptcat*

To see the results of the command view [Figure 5.1](#). No instances of cryptcat.exe or any similarly named file was found on the system, suggesting that encrypted Netcat variants were not used. Instead, the attacker used the plain-text version of nc.exe already confirmed in C:\inetpub\ftproot\ (See [Figure 5.2](#)).

Next I investigated rundll32.exe. The legitimate Windows binary was located at the proper directory `C:\Windows\System32\rundll32.exe`. See figure [Figure 5.3](#) for a screenshot of the directory. While this matches the expected directory for a clean Windows XP installation, it is a dual-use binary often exploited by attackers. It can be used to execute DLL payloads such as Poison Ivy or Netcat, so it is important to continue monitoring processes associated with rundll32 executions.

Incident Explanation

Based on the forensic evidence collected during the lab, the system appears to be the victim of a multi-stage attack designed to gain remote control and steal sensitive information. The attacker seems to have exploited the Windows XP FTP service as its initial infection vector. This allowed them to upload malicious external files into the public directory (`C:\inetpub\ftproot\`). The tools planted by the attacker include:

- **nc.exe (Netcat)** – a command-line utility used as a backdoor to open remote shells.
- **winvnc4.exe (RealVNC server)** – Provides remote access to the desktop without user consent.
- **poisonivy.exe (RAT)** – Allows for persistent command-and-control communications, data exfiltration, and keylogging.
- **runasspc.exe and lock.bat** – Used to escalate privileges and automatically restart the malicious processes after rebooting.

After uploading these malicious files through FTP, the attacker seems to have remotely executed them. Following the installation, the malware set up multiple listening ports (notably 666, 4440, 15152, and 5900) for redundancy and maintaining access. These listening ports allow the attacker to use text-based command shells (Netcat) and graphical sessions (VNC) to remotely interact with the system.

Furthermore, browser artifacts like tracking cookies (ad.wsod, atdmt, scorecardresearch) were found. This suggests that more spyware or adware may have been planted. This provides the attacker with passive user surveillance and activity logging.

Ultimately, the system's normal operations were hijacked to act as both a remote-controlled host and a data-collection point. The attacker turned the XP machine into a staging server and persistent backdoor. The attacker was capable of file transfers, remote command execution, and credential theft.

Future Recommendations & Conclusion

The investigation found that the system was fully compromised by a remote attacker. This lab emphasized the severe security, privacy, and operational risks associated with such incidents. Those risks include data theft, unauthorized remote control, and potential network pivoting to other systems.

Going forward, similar incidents can be prevented by following defensive plans and best practices. You should harden the system by disabling unnecessary services, removing anonymous FTP access, and applying all available security patches. You should also improve network security by implementing a firewall to block non-standard ports such as those found in the lab, and monitor inbound/outbound connections with tools like an IDS. User and credential protection is also vital. Disabling default accounts, using strong passwords, and requiring multi-factor authentication for remote or administrative access are all best practices. If an incident does occur again, it's essential to maintain forensic readiness to reduce risk.

In closing, this lab demonstrated how in a matter of seconds a misconfigured or vulnerable system can be compromised. The attacker gained complete system control without having to face any major challenges through a simple FTP upload. This singular vulnerability allowed the attacker to install multiple backdoors and steal sensitive user data. However, proper forensic investigation and containment steps allows for a successful recovery. This lab demonstrated the importance of secure configurations, continuous monitoring, and proactive security measures.

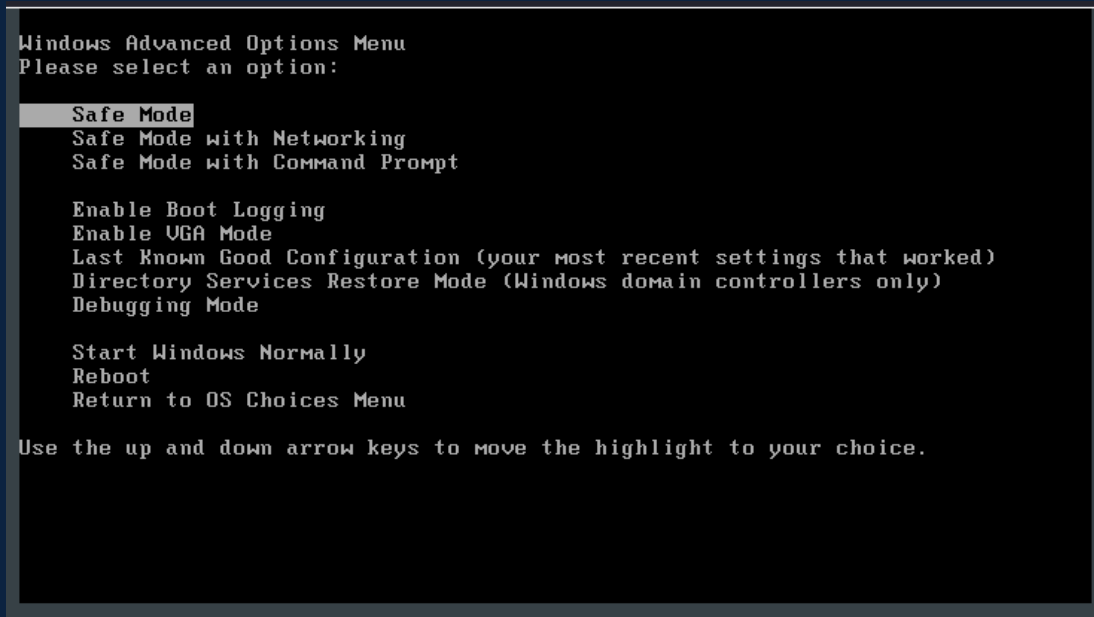
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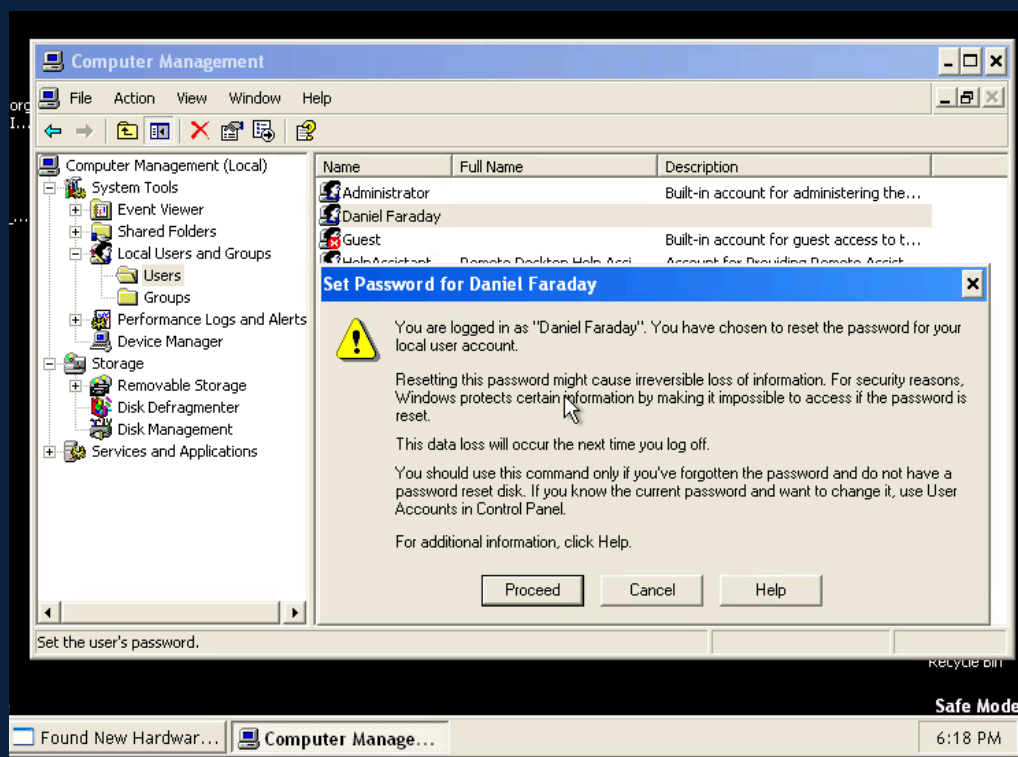
Appendix

1. Gaining Access

- Figure 1.1: Screenshot of the Windows XP Device Booting in Safe Mode.

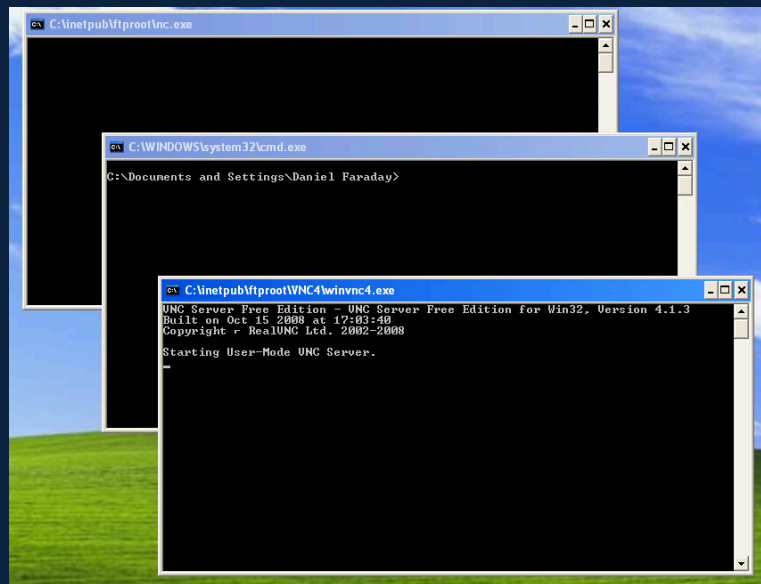


- Figure 1.2: Screenshot of Password Being Reset.

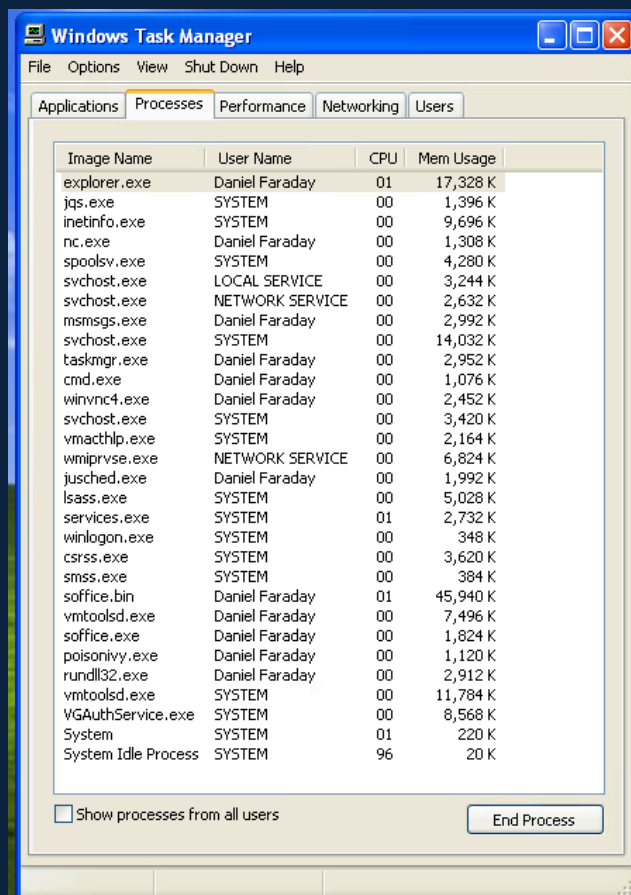


2. System Findings

- Figure 2.1: Screenshot of Suspicious Startup Processes.



- Figure 2.2: Screenshot of Windows Task Manager Process Listing.



3. Network Findings

- Figure 3.1: Screenshot of all Active Connections

```
C:\ Command Prompt
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\Daniel Faraday>netstat -ano

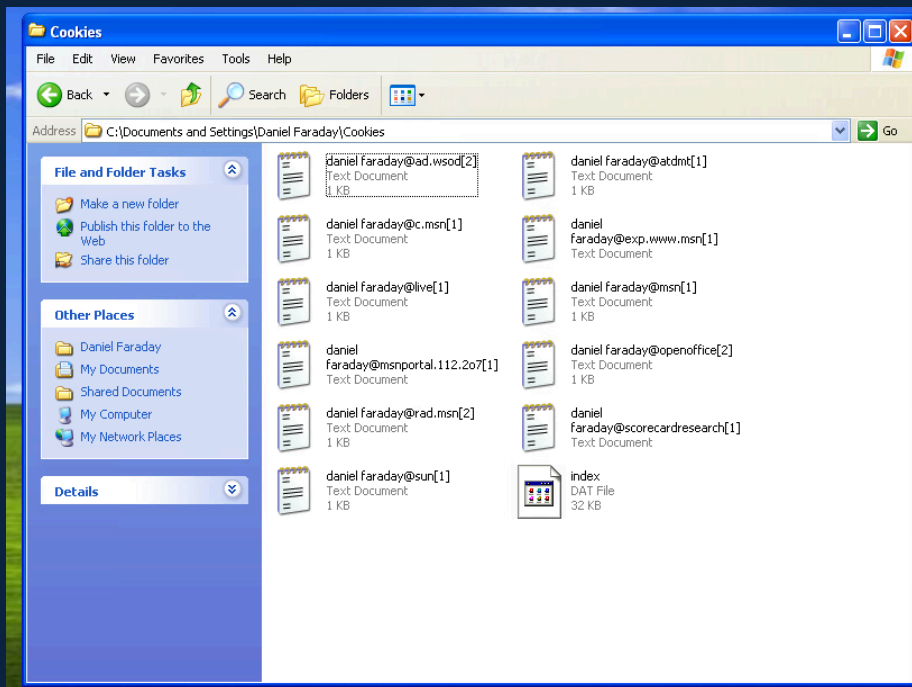
Active Connections

Proto Local Address          Foreign Address         State       PID
TCP    0.0.0.0:21              0.0.0.0:0               LISTENING   1996
TCP    0.0.0.0:25              0.0.0.0:0               LISTENING   1996
TCP    0.0.0.0:80              0.0.0.0:0               LISTENING   1996
TCP    0.0.0.0:135             0.0.0.0:0               LISTENING   1088
TCP    0.0.0.0:443             0.0.0.0:0               LISTENING   1996
TCP    0.0.0.0:445             0.0.0.0:0               LISTENING    4
TCP    0.0.0.0:666             0.0.0.0:0               LISTENING   188
TCP    0.0.0.0:1025            0.0.0.0:0               LISTENING  1340
TCP    0.0.0.0:1029            0.0.0.0:0               LISTENING   1996
TCP    0.0.0.0:4400            0.0.0.0:0               LISTENING   192
TCP    0.0.0.0:5000            0.0.0.0:0               LISTENING  1644
TCP    10.10.0.229:139         0.0.0.0:0               LISTENING    4
TCP    127.0.0.1:5152          0.0.0.0:0               LISTENING  2008
TCP    172.16.3.225:139       0.0.0.0:0               LISTENING    4
UDP    0.0.0.0:135             *:*:                    1088
UDP    0.0.0.0:445             *:*:                    4
UDP    0.0.0.0:5000            *:*:                    876
UDP    0.0.0.0:1026            *:*:                    1604
UDP    0.0.0.0:1033            *:*:                    1996
UDP    0.0.0.0:1038            *:*:                    1340
UDP    0.0.0.0:3456            *:*:                    1996
UDP    10.10.0.229:123         *:*:                    1340
UDP    10.10.0.229:137         *:*:                    4
UDP    10.10.0.229:138         *:*:                    4
UDP    10.10.0.229:1900        *:*:                    1644
UDP    127.0.0.1:123           *:*:                    1340
UDP    127.0.0.1:1030          *:*:                    552
UDP    127.0.0.1:1042          *:*:                    2032
UDP    127.0.0.1:1900          *:*:                    1644
UDP    127.0.0.1:46789         *:*:                    192
UDP    172.16.3.225:123        *:*:                    1340
UDP    172.16.3.225:137        *:*:                    4
UDP    172.16.3.225:138        *:*:                    4
UDP    172.16.3.225:1900       *:*:                    1644

C:\Documents and Settings\Daniel Faraday>
```

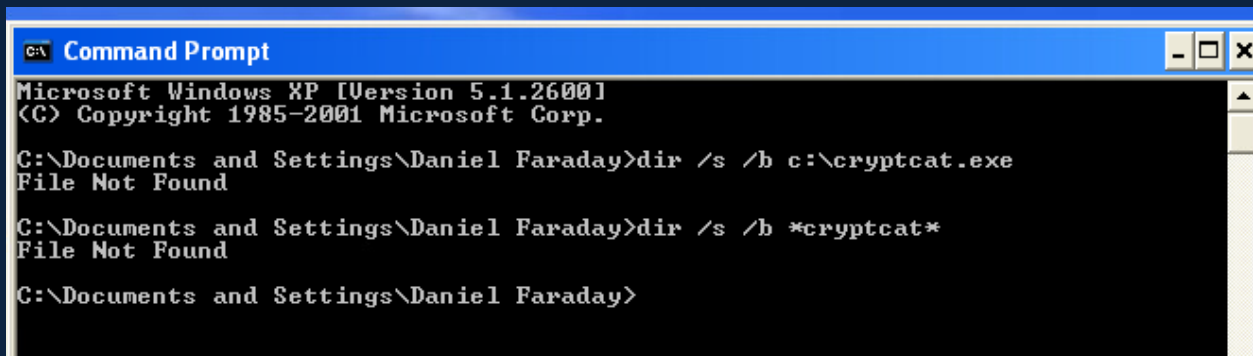
4. Browser Artifacts and Spyware Findings

- Figure 4.1: Screenshot of the Directory for Cookies

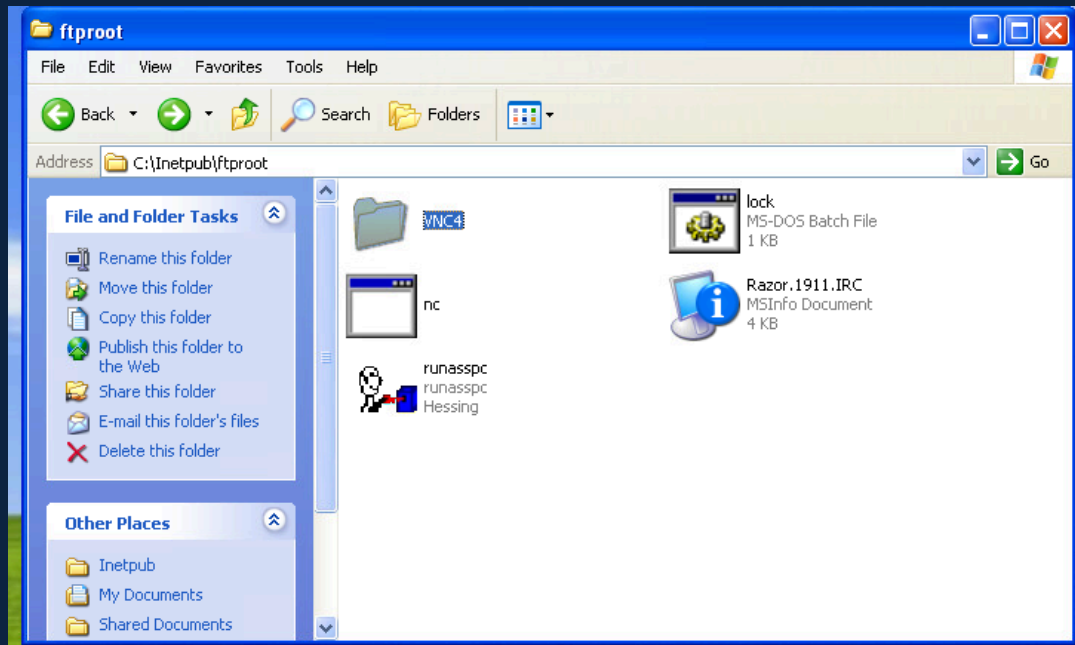


6. Additional Program Findings

- Figure 5.1: Screenshot of Search for Cryptcat



- **Figure 5.2:** Screenshot of *C:\inetpub\ftproot*



- **Figure 5.3:** Screenshot of *rundll32.exe*

